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SMART SWARM ROBOTIC SYSTEM FOR EARTHQUAKE SEARCH AND RESCUE

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Chapter1:Introduction :

1-1 Project Over View and Background :

Earthquakes can cause severe structural damage, making search and rescue operations difficult and dangerous. Mobile robots can support these operations by exploring hazardous environments, detecting obstacles, and collecting information without directly exposing rescuers to every risk.

Using multiple robots can improve exploration by distributing tasks and sharing environmental information. Previous studies have investigated deep-learning-based survivor detection, collaborative mapping, and multi-robot exploration strategies [1]–[3].

This project proposes a simulation-based search and rescue system consisting of two ground robots. The robots will explore an initially unknown environment using simulated LiDAR, construct a shared occupancy grid map, coordinate exploration targets, and calculate a route to a predefined virtual missing-person location.

1-2 Problem Statement and Practical/Academic Significance :

Exploring unknown environments after earthquakes is challenging because of obstacles, limited visibility, and the risk of incomplete or repeated searches. This project addresses these challenges through cooperative exploration, shared mapping, and path planning in a simulated environment.

Academic and Practical Significance:

- **Improved Search Efficiency:** Coordinating two robots can reduce redundant exploration and improve environmental coverage.

- **Enhanced Situational Awareness**: Shared mapping allows robots to combine environmental information and build a more complete representation of the damaged area.
- **Autonomous Navigation:** Integrating frontier-based exploration with A* path planning supports systematic exploration and route planning.
- **Simulation-Based Evaluation:** Testing the integrated system in a simulated environment provides a basis for evaluating its performance before potential real-world implementation.

1-3 Main and Sub Project Objectives :

- The main objective of this project is to develop and simulate a cooperative system of two ground robots capable of exploring earthquake-damaged environments, locating a virtual target, and planning a rescue route.
- Inter-Robot Communication: Develop a communication mechanism that enables the two robots to exchange information, coordinate exploration tasks, share environmental data, and reduce redundant searching.

Sub-Objectives / Technical Requirements:

- **Environment Modeling:** Simulate a damaged building with obstacles.
- **Mapping and Exploration:** Implement LiDAR-based mapping and frontier-based exploration

- **Robot Coordination:** Apply greedy task allocation and map fusion.
- **Path Planning:** Use A* for navigation and replanning.
- **Target Localization:** Simulate target detection and rescue-route planning.

- **Performance Evaluation:** Assess exploration coverage and system performance.

1-4 Basic Definitions and Concepts:

- **Multi-Robot System (MRS):** A system in which multiple robots cooperate to accomplish a shared task.
- **LiDAR (Light Detection and Ranging):** A sensor that measures distances to detect obstacles and surrounding structures.
- **ROS 2 (Robot Operating System):** A flexible framework for writing robot software, providing services designed across a heterogeneous cluster, including hardware abstraction, low-level device control, and message-passing implementations.
- **Occupancy Grid Mapping**: A method of representing an environment using cells classified as occupied, free, or unknown.
- **Frontier-Based Exploration:** An exploration method that identifies boundaries between known and unknown areas.
- **A (A-Star) Algorithm:** * A path-planning algorithm used to find a suitable route between two locations while avoiding obstacles.
- **Map Fusion:** The process of combining maps generated by multiple robots into a shared map

1-5 Project Scope and Technical Requirements :

- **Project Scope:** The project focuses on simulating two ground robots in a two-dimensional earthquake-damaged environment, including mapping, cooperative exploration, path planning, and virtual target localization. It excludes physical robot deployment and real-world disaster testing.
- **Technical Requirements:** The system must support simulated LiDAR sensing, occupancy grid mapping, robot coordination, map fusion, obstacle avoidance, and rescue-route planning. Robot poses are provided by the simulator, and target detection is simulated using a distance threshold.
- The system includes inter-robot communication to support information exchange, exploration coordination, and collaborative mapping. The communication mechanism will be evaluated through simulation to assess its role in improving cooperation between the robots.

1-6 Project Contribution to Knowledge and Field :

This project contributes to the field of search and rescue robotics by proposing an integrated simulation framework that combines cooperative exploration, occupancy grid mapping, map fusion, and autonomous path planning. It provides a test environment for evaluating multi-robot coordination and exploration efficiency in earthquake-damaged buildings, supporting the development of robotic solutions for future disaster response applications.

1-7 Adopted Design Methodology :

The project follows a systematic design methodology consisting of four main phases:

1. **Requirement Analysis and Literature Review:** Defining system requirements and reviewing existing multi-robot search and rescue approaches.
2. **Environment and System Design:** Developing the simulated earthquake-damaged environment and configuring the two ground robots and their sensors.
3. **Simulation and Implementation:** Implementing mapping, frontier-based exploration, task allocation, communication, map fusion, and A* path planning.
4. **Testing, Evaluation, and Analysis:** Evaluating exploration coverage, redundant searching, mapping performance, and rescue-route planning under simulated obstacles.

1-8 Report Structure :

This report is organized into five chapters:

- **Chapter 1** introduces the project background, problem statement, objectives, scope, and methodology.
- **Chapter 2** reviews related studies on multi-robot exploration, collaborative mapping, and robotic search and rescue systems.
- **Chapter 3** describes the proposed system architecture, robot coordination, mapping, and navigation methods
- **Chapter 4** presents the implementation, simulation results, and performance evaluation.
- **Chapter 5** summarizes the conclusions, project limitations, and recommendations for future work.

1-9 Conclusion :

This chapter established the background, objectives, scope, and methodology of the proposed multi-robot search and rescue system. The following chapter reviews related studies and identifies the research gap addressed by this project.

Chapter 2: Literature Review and Previous Studies:

2-1 Previous Studies in the Project Field

Recent studies have explored robotic technologies for search and rescue operations, focusing on survivor detection, multi-robot exploration, and collaborative mapping. The following studies provide the theoretical and technical basis for the proposed project.

1. Survivor Detection Using Deep Learning (Jadeja et al., 2024):

This study investigates survivor detection using a snake robot and deep learning algorithms, including Faster R-CNN, SSD, YOLOv8, and YOLOv10. The results demonstrate the effectiveness of YOLOv10 in detecting survivors in images of earthquake debris. However, the study primarily focuses on visual detection rather than cooperative exploration by multiple robots.

2. Distributed Multi-Robot Exploration and Mapping (Pongsirijinda et al., 2024):

This study proposes DMPF-Explore, a distributed exploration framework combining submap-based collaborative mapping with potential-field-based exploration. Its methods improve map consistency, frontier detection, and cooperation between robots. These concepts are relevant to the proposed system's collaborative mapping and exploration.

3. Reinforcement Learning-Based Multi-Robot Exploration (Calzolari et al., 2024):

This study presents a decentralized multi-robot exploration framework using reinforcement learning, explicit communication, and density-based frontier search. It also uses A*-based trajectory information to support exploration decisions and reduce redundant searching. These

2-2 Comparative Analysis of Existing Systems and Products

Existing research presents different approaches to robotic search and rescue, including deep learning-based detection, distributed exploration, and communication-based coordination. Table 2.1 compares the selected studies in terms of their main techniques, strengths, and limitations.

Table 2.1: Comparative Analysis of Existing Approaches

Study	Main Techniques	Strengths	Limitations	Use in Our Project
Jadeja et al. (2024)	Deep learning, YOLO, SSD, Faster R-CNN	Supports automated survivor detection in disaster environments.	Focuses primarily on detection rather than cooperative exploration.	Provides background on survivor detection. Our project instead focuses on cooperative exploration and simulated target localization, without deep-learning-based visual detection.
Pongsirijinda et al. (2024)	Distributed exploration, potential fields, submap-based mapping	Supports collaborative exploration and map integration.	Focuses on exploration and mapping rather than the complete rescue process.	Informs our use of cooperative exploration and shared-map concepts for two simulated ground robots.
Calzolari et al. (2024)	Reinforcement learning, explicit communication, frontier search, A*	Supports coordinated exploration and information sharing between robots.	Focuses on exploration rather than complete search-and-rescue operations.	Provides ideas about robot communication, frontier-based exploration, and A*-based path planning. Our project uses a simpler simulation-based coordination approach.

2-3 Research Gap and Proposed Innovation

- The reviewed studies address important aspects of robotic search and rescue, including survivor detection, multi-robot exploration, and collaborative mapping. However, the proposed project focuses on integrating cooperative exploration, shared mapping, task allocation, and rescue-route planning within a single simulation framework. Its main contribution is evaluating how these components work together using two ground robots in a simulated earthquake-damaged environment.

- Another important aspect of the proposed system is the integration of inter-robot communication with cooperative exploration, task allocation, and shared mapping. This integration aims to enable the robots to exchange relevant information and coordinate their actions within a unified simulation framework. The project will evaluate how communication supports cooperation and reduces redundant exploration during search and rescue operations

2-4 Theoretical Framework and Reference Studies

The project is based on multi-robot coordination, occupancy grid mapping, frontier-based exploration, greedy task allocation, and A* path planning. These methods are integrated to support autonomous navigation, information sharing, and rescue-route planning. The system is developed and evaluated in a two-dimensional simulation using virtual robots and sensors.

Communication in Multi-Robot Systems:

• **Role of Communication:** Enable information exchange and coordinated decision-making among multiple robots.

• **Shared Environmental Representation:** Use exchanged information to support collaborative mapping and maintain a more complete representation of the environment.

• **Cooperative Exploration:** Coordinate robot movements and exploration tasks to reduce overlap and improve search efficiency.

• **Communication Mechanisms:** Consider suitable technologies, such as ROS 2 topics and DDS, depending on the selected simulation framework.

• **Key Design Considerations:** Address software compatibility, communication reliability, data-exjchange requirements, and their effects on cooperative performance.

Technical Specifications

Smart Swarm Robotic System for Earthquake Search and Rescue

No.	Technical Specification	Description
1	Number of Robots	Two simulated ground robots
2	Simulation Environment	Two-dimensional damaged environment
3	Simulation Area	Approximately 100–400 m²
4	Robot Dimensions	Length: 0.10–0.15 m; width: 0.08–0.12 m
5	Number and Type of Wheels	Three wheels per robot: two independently motorized drive wheels and one free wheel for balance
6	Robot Weight	Target weight: not exceeding 200 g (preliminary design value)
7	Robot Steering Method	Differential Drive
8	Distance Between Robots	Variable according to task allocation and environment layout
9	Linear Speed	Approximately 0.1–0.5 m/s
10	LiDAR Sensor	Field of view up to 360°; approximate range of 3–10 m
11	Map Resolution	Grid cell size of approximately 0.10 m (10 cm), subject to simulation testing
12	Environment Exploration	Target coverage of 80–90% of the accessible area
13	Task Allocation	Assign exploration targets to reduce redundant exploration
14	Path Planning Algorithm	A*
15	Cooperation Algorithm	Greedy Task Allocation
16	Communication Between Robots	ROS 2 for exchanging maps, robot status, and exploration targets
17	Control and Coordination Framework	Open-RMF
18	Operating Time	To be determined according to simulation duration and test scenarios
19	Battery	Not required for simulation; to be specified if a physical robot is selected
20	Performance Evaluation	Measure exploration coverage, exploration time, redundant exploration, and path-planning success

Note: Numerical ranges and dimensions are preliminary design estimates and may be adjusted according to the selected simulation environment and robot model.

References

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